Prediction of PVT Behaviour, Enthalpy, Entropy and Fugacity of Real Gases Using Cubic Equations of State

Computer Project — Chemical Engineering Thermodynamics (Prof. S. K. Gupta)
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1 Objective

To develop a program that, for a pure gas at given temperature T and pressure P, predicts (i) the PVT behaviour (compressibility factor Z and molar volume V), (ii) the enthalpy H and entropy S, and (iii) the pure-component fugacity f (computer project), using cubic equations of state (EOS), and to check the predictions against accurate reference data. Four cubic EOS are implemented — van der Waals (vdW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR) — through one generic form, so that their accuracy can be compared.

2 Theory

Generic cubic EOS. All four equations are special cases of

$$P = \frac{RT}{V-b} - \frac{a(T)}{(V+\varepsilon b)(V+\sigma b)}, \qquad a(T) = \Psi\,\frac{\alpha(T_r,\omega)R^2T_c^2}{P_c}, \qquad b = \Omega\,\frac{RT_c}{P_c} \tag{1}$$

with the constants of Table 1. Only T_c, P_c and the acentric factor ω of the gas are needed.

Table 1: Parameters of the cubic equations of state (Smith, Van Ness & Abbott).

EOS	σ	ε	Ω	Ψ	$\alpha(T_r)$
vdW	0	0	1/8	27/64	1
RK	1	0	0.08664	0.42748	$T_r^{-1/2}$
SRK	1	0	0.08664	0.42748	$[1+m(1-T_r^{1/2})]^2$, $m = 0.480 + 1.574\omega - 0.176\omega^2$
PR	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	$[1+\kappa(1-T_r^{1/2})]^2$, $\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$

PVT — cubic in Z. With $\beta = bP/RT$ and $q = a/bRT$, substituting $V = ZRT/P$ gives

$$(Z-\beta)(Z+\varepsilon\beta)(Z+\sigma\beta) - (Z+\varepsilon\beta)(Z+\sigma\beta) + q\beta(Z-\beta) = 0. \tag{2}$$

For PR this is $Z^3 - (1-B)Z^2 + (A - 3B^2 - 2B)Z - (AB - B^2 - B^3) = 0$ with $A = aP/(RT)^2$, $B = \beta$. All three roots are found (`numpy.roots`); only real roots with $Z > \beta$ are physical. The largest root is the vapour, the smallest the liquid. When three real roots exist and no phase is specified, the program picks the root with the lower fugacity (lower Gibbs energy), i.e. the stable phase. Then $V = ZRT/P$.

H and S — residual properties. Defining $I = \frac{1}{\sigma-\varepsilon}\ln\frac{Z+\sigma\beta}{Z+\varepsilon\beta}$ ($I = \beta/Z$ for vdW), the departures from the ideal gas at the same T and P follow from $M^R = M - M^{ig}$:

$$\frac{H^R}{RT} = Z - 1 + \left[\frac{d\ln\alpha}{d\ln T_r} - 1\right] qI, \qquad \frac{S^R}{R} = \ln(Z-\beta) + \frac{d\ln\alpha}{d\ln T_r}\,qI, \tag{3}$$

where $d\ln\alpha/d\ln T_r = 0$ (vdW), $-\frac{1}{2}$ (RK) and $-m\sqrt{T_r}/[1+m(1-\sqrt{T_r})]$ (SRK); for PR the same expression with m replaced by κ. Taking the ideal gas at $T_0 = 298.15\,\mathrm{K}$, $P_0 = 1\,\mathrm{bar}$ as the zero of H and S, with $C_P^{ig}/R = A + BT + CT^2 + DT^{-2}$,

$$H = \int_{T_0}^{T} C_P^{ig}\,dT + H^R, \qquad S = \int_{T_0}^{T}\frac{C_P^{ig}}{T}\,dT - R\ln\frac{P}{P_0} + S^R. \tag{4}$$

Fugacity. From $\ln\phi = G^R/RT = \int_0^P (Z-1)\,dP/P$ at constant T:

$$\ln\phi = Z - 1 - \ln(Z-\beta) - qI, \qquad f = \phi P. \tag{5}$$

Saturation pressure (extension). Below T_c, vapour–liquid equilibrium requires $f^L = f^V$. Starting from the Wilson estimate, P is updated as $P_{\mathrm{new}} = P\,\phi^L/\phi^V$ until $|\phi^L/\phi^V - 1| < 10^{-10}$.

3 Program structure

The program is written in Python (NumPy only for calculation) and organised in modules, kept in the `python/` folder of the repository:

- `data.py` — T_c, P_c, ω, C_P^{ig} coefficients for 9 gases (SVNA App. B, C).
- `eos.py` — EOS parameters, $\alpha(T_r)$, cubic solver and root selection, H^R, S^R, $\ln\phi$, ideal-gas H and S, saturation pressure.
- `main.py` — interactive menu (properties at T, P; saturation pressure; comparison of all EOS) and command line, e.g. `python main.py methane 250 50 PR`.
- `validate.py`, `make_plots.py` — comparison with reference data and figures.
- `index.html` — browser version of the same program (JavaScript port giving identical results), live at `https://krishnakr9861.github.io/CHL2002-PROJECT/`.

Algorithm: read gas, T, P, EOS → compute a, b, α, $d\ln\alpha/d\ln T_r$ → form β, q and solve the cubic → select root → compute V, H^R, S^R, ϕ → add ideal-gas parts to get H, S → print. For methane at 250 K and 50 bar (PR) the program gives $Z = 0.8117$, $V = 337.5\,\mathrm{cm^3/mol}$, $H^R = -1295\,\mathrm{J/mol}$, $S^R = -3.59\,\mathrm{J/mol\,K}$, $H = -2908\,\mathrm{J/mol}$, $S = -42.01\,\mathrm{J/mol\,K}$, $\phi = 0.826$, $f = 41.3\,\mathrm{bar}$.

Checks. (1) The SVNA worked example (n-butane, 350 K, 9.457 bar) was recomputed: $Z^V = 0.8667$, 0.8305, 0.8191, 0.8081 for vdW, RK, SRK, PR, in agreement with the textbook values to within about 0.3 %; small differences can come from rounding of T_c, P_c and ω. (2) Thermodynamic consistency: $H^R/RT - S^R/R = \ln\phi$ holds to machine precision, and H^R equals $-RT^2(\partial\ln\phi/\partial T)_P$ evaluated numerically. (3) As $P \to 0$: $Z \to 1$, $\phi \to 1$, $H^R, S^R \to 0$.

4 Results and validation

Reference values were obtained from CoolProp, which implements the high-accuracy multiparameter equations of state used by the NIST Chemistry WebBook. Departure functions are compared, so the choice of reference state does not matter.

Table 2: Peng–Robinson predictions vs reference data (single-phase gas / supercritical states).

Gas	T (K)	P (bar)	Z	Z_{ref}	H^R (J/mol)	H^R_{ref} (J/mol)	S^R (J/mol K)	S^R_{ref} (J/mol K)	ϕ	ϕ_{ref}
Methane	250.0	50	0.812	0.836	−1295	−1185	−3.59	−3.40	0.826	0.851
Methane	300.0	100	0.834	0.856	−1759	−1604	−4.24	−4.02	0.823	0.853
Nitrogen	200.0	100	0.824	0.845	−1480	−1376	−5.47	−5.28	0.793	0.825
Nitrogen	300.0	200	1.028	1.057	−1016	−905	−3.18	−3.10	0.976	1.010
CO_2	320.0	50	0.754	0.771	−2210	−2139	−4.98	−4.93	0.794	0.810
CO_2	350.0	100	0.651	0.661	−4014	−3830	−8.63	−8.34	0.711	0.731
Propane	400.0	30	0.767	0.787	−2534	−2416	−4.49	−4.39	0.801	0.820
Water	573.15	50	0.862	0.857	−2069	−2725	−2.51	−3.67	0.877	0.878

Table 3: Mean absolute % error, 8 states of Table 2.

EOS	Z	H^R	S^R	ϕ
vdW	4.2	20.8	25.9	3.3
RK	1.3	8.9	12.9	1.2
SRK	1.6	5.4	6.4	0.9
PR	2.2	9.5	6.9	2.6

Table 4: Saturation pressure (bar) from $f^L = f^V$.

Gas	T (K)	Ref.	vdW	RK	SRK	PR
Propane	300	9.98	17.37	11.53	10.10	9.99
CO_2	280	41.61	52.54	45.59	41.95	41.56
Water	373.15	1.014	15.18	2.94	0.924	0.960
Water	473.15	15.55	55.02	27.57	15.68	15.58
Methane	150	10.40	16.34	10.05	10.49	10.45
Ammonia	300	10.61	29.74	15.41	10.63	10.53

5 Discussion

- **PVT.** The three two-parameter EOS with a temperature-dependent α (or $T_r^{-1/2}$) predict Z of non-polar gases within about 1–3 %. The vdW equation is the least accurate (4 % mean error) and its Z curve is distorted near the critical region (Fig. 1). PR reproduces the minimum in the Z–P isotherms of methane (Fig. 2a), and all four EOS show it for CO_2 (Fig. 1); the largest deviations occur near T_c and at high density, where a two-parameter cubic cannot capture the true volume.
- **H and S.** Residual properties are more sensitive than Z because they depend on $d\alpha/dT$. SRK gives the lowest errors in both H^R and S^R; PR is close to SRK for S^R but not for H^R, where RK is comparable (Table 3). Water strongly affects these averages: without it, the mean H^R errors are 4.4, 2.6 and 7.4 % for RK, SRK and PR, and the S^R errors 7.8, 2.9 and 3.4 %; vdW, with $\alpha = 1$, misses most of the temperature dependence of attraction, so its H^R and S^R level off at high pressure (Fig. 1). Errors in H^R are larger in percentage terms when H^R itself is small (e.g. N_2 at 300 K, 200 bar).
- **Fugacity.** ϕ is predicted within about 1–4 % for all gases. Since $\ln\phi$ integrates $(Z-1)/P$, errors in Z partly

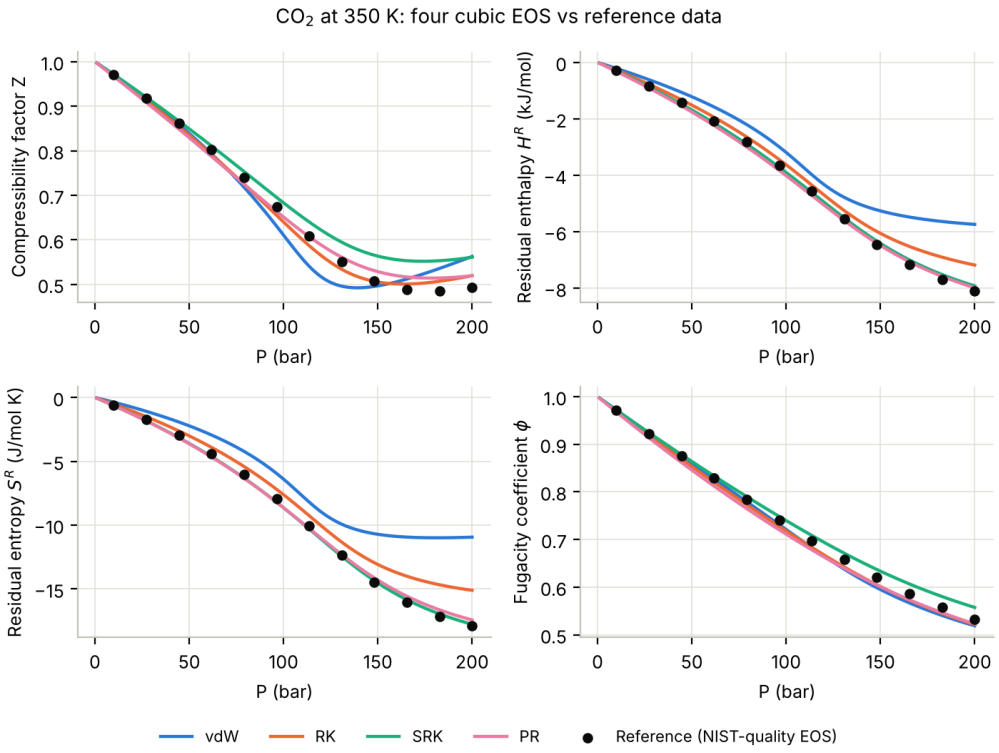


Figure 1: Z, H^R, S^R and ϕ of CO_2 at 350 K ($T_r = 1.15$) from the four EOS, with reference data (dots).

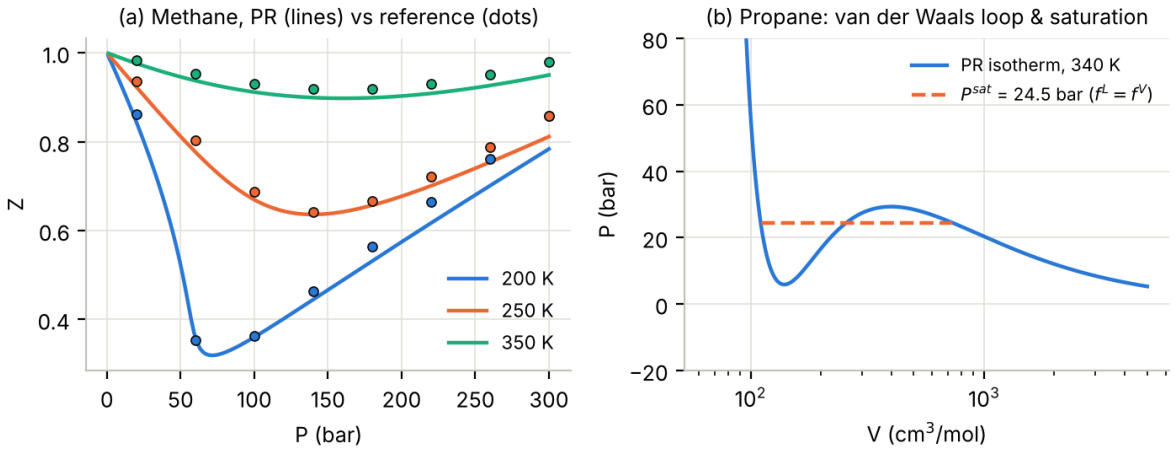


Figure 2: (a) Z–P isotherms of methane by PR with reference data. (b) Subcritical PR isotherm of propane at 340 K showing the van der Waals loop; the saturation pressure where $f^L = f^V$ is marked.

cancel. At high pressure, ϕ falls well below 1 (CO_2 at 350 K: $\phi = 0.71$, so $f \approx 71$ bar at 100 bar) — the ideal-gas assumption $f = P$ would be badly wrong.

- **Saturation pressure.** Because $\alpha(T_r, \omega)$ in SRK and PR is fitted to vapour pressure, these EOS predict P^{sat} within $\approx 1\,\%$ for the non-polar fluids and for ammonia, but water at 373 K ($T_r = 0.58$) is under-predicted by 9 % (SRK) and 5 % (PR). PR is the most accurate in most cases, though SRK is closer for ammonia (Table 4). vdW and RK overpredict P^{sat} strongly. This shows the main practical use of fugacity: phase equilibrium.
- **Limitations.** Water and ammonia are polar and hydrogen-bonded; for water at 573 K, 50 bar (close to saturation, $P^{sat} = 86$ bar) PR gives Z and ϕ well but H^R and S^R are 24–32 % low. The simple C_P^{ig} polynomials are valid only over their fitted range (typically 298–1500 K).

6 Conclusion

A single program based on the generic cubic EOS was developed that calculates Z, V, H, S, ϕ and f of nine pure gases with four equations of state, and additionally finds the saturation pressure from the equal-fugacity condition. The results agree with the SVNA textbook example, satisfy thermodynamic consistency checks, and agree with reference data within a few per cent for non-polar gases. For gas-phase Z and ϕ, RK, SRK and PR perform similarly on this small set of 8 states (RK lowest); SRK gives the best H^R and S^R. PR is recommended for general use because it gives the best vapour pressures and the same form serves both phases, while vdW is suitable only for qualitative behaviour.

References. [1] J. M. Smith, H. C. Van Ness, M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., McGraw-Hill, Ch. 3, 6, 11 and App. B, C. [2] D.-Y. Peng, D. B. Robinson, *Ind. Eng. Chem. Fundam.* 15 (1976) 59–64. [3] G. Soave, *Chem. Eng. Sci.* 27 (1972) 1197–1203. [4] I. H. Bell et al., CoolProp, *Ind. Eng. Chem. Res.* 53 (2014) 2498–2508.